Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 5 times with different random seeds. All experiments were run on a single GPU with 24 GB of memory.

2 Results

Figure 1 shows the f1 score by dataset.

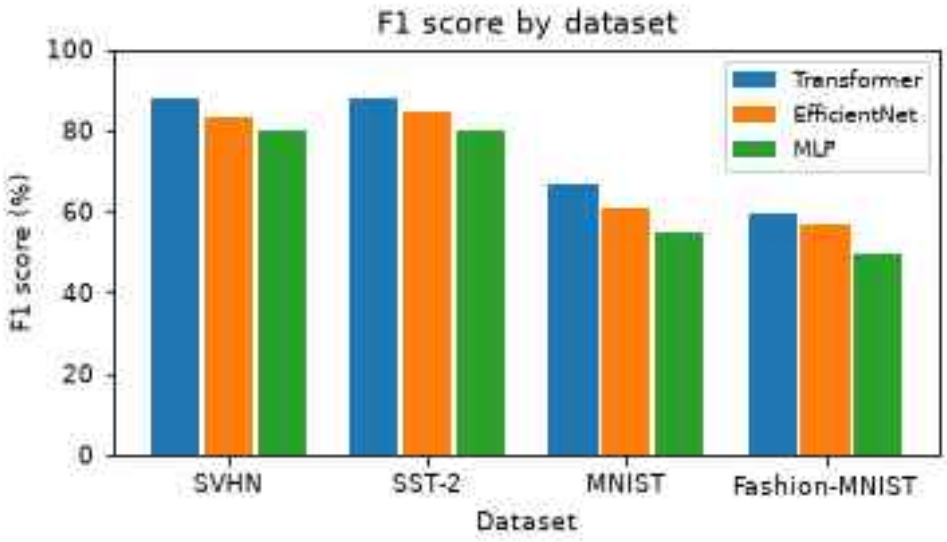


Figure 1: F1 score by dataset for the models in this experiment.

On Fashion-MNIST, EfficientNet is 7 points better than MLP (Figure 1). Transformer achieves about 60% F1 score on Fashion-MNIST. Transformer outperforms MLP by 58.1 percentage points on SST-2.

Figure 2 shows the accuracy by dataset.

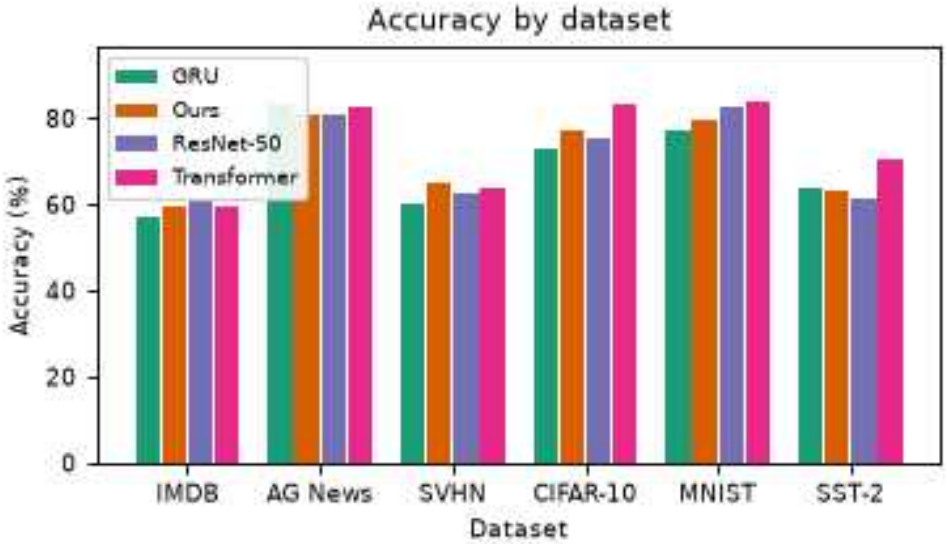


Figure 2: Accuracy by dataset for the models in this experiment.

GRU reaches 63.9% accuracy on SST-2 (Figure 2). Transformer outperforms ResNet-50 by 10.8 percentage points on MNIST.

Figure 3 shows the validation accuracy during training.

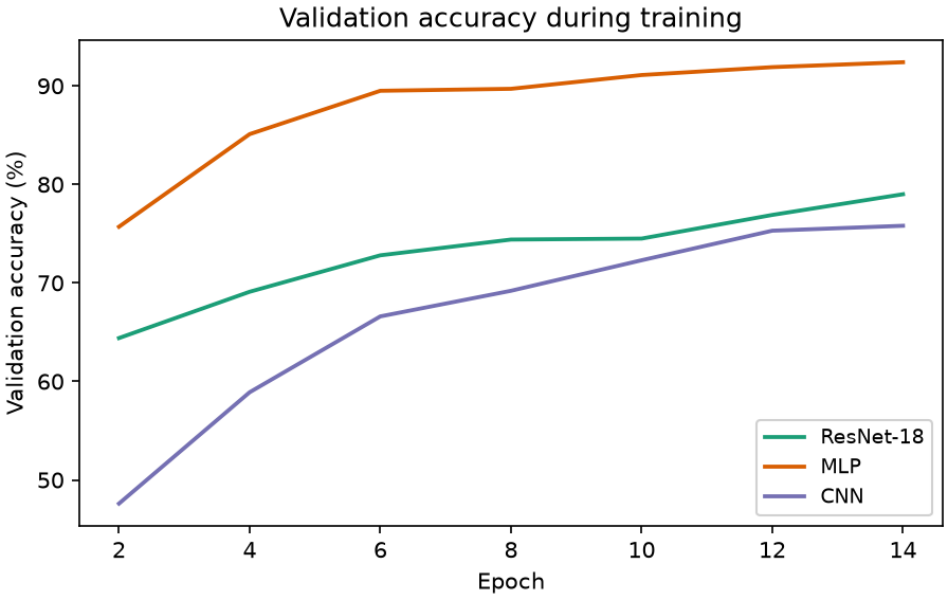


Figure 3: Validation accuracy during training for the models in this experiment.

As shown in Figure 3, from epoch 12 to epoch 14, the validation accuracy of CNN improves by a relative 2.0%. MLP reaches 94.6% validation accuracy at epoch 8 (Figure 3). As shown in Figure 3, MLP attains a validation accuracy of 96.8% at epoch 12. MLP attains a validation accuracy of 84.6% at epoch 6.

3 Discussion

Logistic Regression reaches 92.6% accuracy on IMDB. Overall, the results are in line with what we expected from the literature.